Year 8
Mathematics

Unit 3: Working with Powers

3.4 Expanding and Factorising Expressions

Expanding and Factorising Expressions

STARTER · 5 MIN

Date: _____ **Title: Expanding and Factorising Expressions**

LEARNING OBJECTIVES

- To be able to expand and factorise algebraic expressions

STARTER *Answer in your book. No calculators.*

1. Simplify $4n \times -5n$
2. Expand $y(y + 3)$
3. Expand $2(3x + 5)$
4. Expand $s(10 - s)$

Starter — Answers

STARTER

Tick or correct in green pen

1. $-20n^2$
2. $y^2 + 3y$
3. $6x + 10$
4. $10s - s^2$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can expand brackets accurately	☐
SILVER	I can factorise expressions by identifying common factors	☐
GOLD	I can link area models to algebraic expressions	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Expanding and Factorising Expressions

INTRODUCTION · 3 MIN

A term outside a bracket multiplies every term inside. Factorising reverses this.

Area model

Area = length × width.

$x(3x + 2) = 3x^2 + 2x$

Expand

Multiply each term, add powers.

$x^2(x + 4) = x^3 + 4x^2$

Factorise

Take out the HCF, numbers and letters.

$15z^2 - 35z = 5z(3z - 7)$

Identity

Expand both sides to check.

$4x^3 + x(3x^2 + 7x) \equiv 7x^2(x + 1)$

Examples 1 and 2 — Expanding with Powers

EXAMPLES · 5 MIN

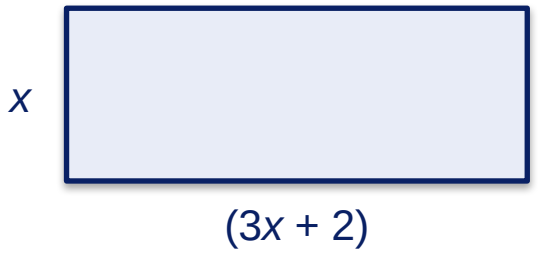
BRONZE

Watch and copy into your book

Hint Multiply every term Add the powers

Example 1

Write the area as an expression, then expand.



Example 2

Expand

$x^2(x + 4)$

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Length × width

Area $= x(3x + 2)$

$\mathbf{= 3x^2 + 2x}$

Tip: $x \times 3x = 3x^2$.

Example 2

Multiply both terms by x^2

$x^2 \times x = x^3$

$x^2 \times 4 = 4x^2$

Answer $= x^3 + 4x^2$

Tip: add the powers.

Task 1

PRACTICE · 9 MIN

BRONZE *Expand*

Hint Multiply every term Add the powers

1. $2y(5 + 3y)$

2. $5z(3z - 7)$

3. $2x^5(7x - 3)$

4. $5x(x^3 + 2x^2 + 7)$

5. $x^2(x^2 - 5x + 7)$

★ $2x(7x^2 + 13)$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 $10y + 6y^2$

2 $15z^2 - 35z$

3 $14x^6 - 6x^5$

4 $5x^4 + 10x^3 + 35x$

5 $x^4 - 5x^3 + 7x^2$

★ $14x^3 + 26x$

Examples 3 and 4 — Factorising with Powers

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint HCF: numbers and letters Lowest power of each letter

Example 3

Factorise completely

$15x^3 - 3x$

Example 4

Factorise completely

$3y^5 + 15y^3$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

HCF: 3 and x

$$15x^3 - 3x = 3x(5x^2 - 1)$$

Tip: $3x \div 3x = 1$, not 0.

Example 4

HCF: 3 and y^3

$$3y^5 + 15y^3 = 3y^3(y^2 + 5)$$

Tip: lowest power of y.

Task 2

PRACTICE · 9 MIN

SILVER *Factorise completely*

Hint HCF: numbers and letters Lowest power of each letter

1. $32x + 16x^2$

2. $15x + 21x^2$

3. $3x^3 + 6x$

4. $y^2 - 7y^4$

5. $12y^4 - 4y^3$

★ $5m^3 + 10m^4$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1 $16x(2 + x)$

2 $3x(5 + 7x)$

3 $3x(x^2 + 2)$

4 $y^2(1 - 7y^2)$

5 $4y^3(3y - 1)$

★ $5m^3(1 + 2m)$

Examples 5 and 6 — Show That and Identities

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint Show every step ≡ : true for all values

Example 5

(3 marks)

Show that

$4x^3 + x(3x^2 + 7x) \equiv 7x^2(x + 1)$

Example 6

(3 marks)

An L-shape is made of three rectangles: $2y$ by $(3y - 1)$, y by $(y - 1)$ and $2y$ by $(y + 3)$.

Show its area is $9y^2 + 3y$.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Expand, collect, then factorise

$4x^3 + 3x^3 + 7x^2 \quad = 7x^3 + 7x^2$

$\mathbf{= 7x^2(x + 1)}$

Tip: 'show that' means show every step.

Example 6

Add the three areas

$6y^2 - 2y + y^2 - y + 2y^2 + 6y$

$\mathbf{Total \quad = 9y^2 + 3y}$

Tip: expand each part first.

Task 3

PRACTICE · 9 MIN

GOLD *Exam-style: expand, factorise, show*

Hint Show every step ≡ : true for all values

1 Expand and simplify
$x(x + 5) + 3(x^2 - 2x)$

2 Expand $3y(2y^2 - 4)$

3 Factorise $9y^3 - 6y^2$

4 Factorise $8p^2q + 12pq$

5 Is $2x(x + 3) \equiv 2x^2 + 3$?
Explain.

★ Show $3(x + 2) + x(x - 3)$
$\equiv x^2 + 6$

Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1. $4x^2 - x$
2. $6y^3 - 12y$
3. $3y^2(3y - 2)$
4. $4pq(2p + 3)$
5. No: $2x^2 + 6x$

★ $3x + 6 + x^2 - 3x = x^2 + 6$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Expand and simplify $3(x + 4) + x(x - 2)$

SILVER Factorise completely $6x^2 + 10x$

GOLD Factorise completely $12y^3 - 8y^2$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $x^2 + x + 12$

SILVER $2x(3x + 5)$

GOLD $4y^2(3y - 2)$

Exam Practice — Expanding

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint Multiply every term Add the powers

Question 1

(2 marks)

Expand

(a) $x(x + 6)$ (1)

.....

(b) $3y^2(2y - 5)$ (1)

.....

Question 2

(2 marks)

Expand and simplify

$x(x + 4) + 2(x^2 - x)$

.....

Exam Practice — Factorising Completely

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint
HCF: numbers and letters
Lowest power of each letter

Question 3

(2 marks)

Factorise completely $14x^3 - 21x^2$

.....

Question 4

(2 marks)

Factorise completely $6a^2b + 9ab^2$

.....

Exam Practice — Show That

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint Show every step ≡ : true for all values

Question 5

(3 marks)

Show that

$5x(x + 2) - 2(x^2 + 5x) \equiv 3x^2$

.....

Question 6

(3 marks)

A rectangle is $2x$ cm by $(x + 5)$ cm.
A square of side x cm is cut out.

Show the area left is $x^2 + 10x$ cm^2.
Then factorise it.

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) $x^2 + 6x$ — B1

(b) $6y^3 - 15y^2$ — B1

Tip: add the powers.

Question 2

$x^2 + 4x + 2x^2 - 2x$ — M1

$3x^2 + 2x$ — A1

Tip: collect like terms.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$7x^2(2x - 3)$ B2

B1 for a partly factorised answer

Tip: HCF $7x^2$.

Question 4

$3ab(2a + 3b)$ B2

B1 for a partly factorised answer

Tip: HCF $3ab$.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$5x^2 + 10x - 2x^2 - 10x$ — M1 M1

$= 3x^2$ — A1

Tip: $-2 \times 5x = -10x$.

Question 6

$2x^2 + 10x - x^2$ — M1

$= x^2 + 10x$ — A1

$x(x + 10)$ — B1

Tip: rectangle area minus square area.